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NCAA Football Players are at Higher Risk of Upper Extremity Injury After First-Time Concussion

Running Title: Concussion Increases UE Injury Risk in NCAA Football Players

Carson H Gardner MD[1]

Jacob L Kotlier BA[1]

Amir Fathi BS[1]

Justin Castonguay BS[1]

Ashley A Thompson BS[1]

Ioanna K Bolia MD[1]

Frank A Petrigliano MD[1]

Joseph N Liu MD[1]

Alexander E Weber MD[1]

Seth C Gamradt MD[1]

[1] *Keck School of Medicine of USC, Department of Orthopaedic Surgery, Los Angeles, CA*

Corresponding Author:

Jacob L Kotlier, BA

USC Department of Orthopaedic Surgery

1520 San Pablo Street, Suite 2000, Los Angeles, CA, 90033

Email: Kotlierja@gmail.com

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Abstract

Background: Previous research has demonstrated that concussions increase the risk of subsequent lower extremity musculoskeletal injury in athletes. However, the risk of upper extremity injury in athletes' post-concussion is poorly understood.

Methods: All concussed football players within a National Collegiate Athletic Association (NCAA) Division I conference athletic database were identified between 2017-2021. After exclusions, each athlete experiencing their first concussion was then retrospectively reviewed for upper extremity injuries in the year prior to their concussion and in the year beginning at 90 days after their concussion. All upper extremity injuries were identified and the odds ratio, 95% confidence interval, and statistical significance between groups were calculated in Microsoft Excel.

Results: 160 de-identified football players from a single conference who were first diagnosed with concussions in the seasons from 2017 through 2021 met inclusion criteria. In these athletes the odds of upper extremity injury in year following first diagnosed concussion were 2.36 times higher than in the year prior (95% CI 1.13 – 4.95, $p = 0.02$). Shoulder was the most common site of injury with 57.7% of injuries compared to 19.2% in the hand, 15.4% in the elbow, 7.7% in the forearm, and 0% in the wrist.

Conclusion: This study demonstrates that collegiate football players are at a 2.36 times greater risk of upper extremity injury in the year following their first diagnosed concussion compared to

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the year preceding it. The most common site of upper extremity injury after concussion was the shoulder.

Level of evidence: III

Key Terms: Head Injuries, Concussion, Football (America), Injury Prevention, General Sports Trauma

Introduction

Concussion is defined at the 6th International Conference on Concussion in Sport as: "Sport-related concussion is a traumatic brain injury caused by a direct blow to the head, neck or body resulting in an impulsive force being transmitted to the brain that occurs in sports and exercise-related activities."[31] While concussions exhibit symptoms in the acute phase, they may also lead to long-term impairments such as headaches, vision problems, vestibular symptoms, fatigue, depression, sleep disorders and behavioral issues.[32] The mainstay of treatment for concussion includes identifying issues amenable to specific treatment, followed by rest, careful monitoring of symptoms and recovery, and submaximal exercise.[32] Typical return to play criteria for athletes sustaining a concussion focuses on resolution of concussion-related symptoms and neurocognitive deficits followed by remaining asymptomatic with exercise.[32]

The impact of concussion on lower extremity musculoskeletal injury risk has been shown in multiple areas of the lower extremity including foot, ankle, knee and hip.[26] Prior studies have attempted to explain and characterize the increased risk of lower extremity musculoskeletal injury after concussion.[3-7,12,17,18,22-24,27,29,30,33,34,36] Notably, athletes with a history of concussion have been shown to have altered gait, deficits in landing biomechanics, and deficiencies in dual-task challenges that combine mechanical and cognitive components.[5,17,18,23,24] Such deficiencies have been proposed to increase risk of lower extremity musculoskeletal injury following

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concussion. Increased lower extremity musculoskeletal injury risk after concussion is apparent in athletes ranging from the high school to professional level, active-duty military personnel, and the general population and higher rate of lower extremity musculoskeletal injury after concussion has been demonstrated to persist as long as three years after the original head injury.[15]

While the risk of lower extremity musculoskeletal injury after concussion and explanations for it have been studied extensively, the link between concussion and upper extremity injury is not well studied. Our review current literature on the topic produced only one paper. After controlling for confounders, a recent study in the military population demonstrated the risk of upper extremity injury to be almost two times greater in the year after returning from a concussion compared to matched controls.[35] Study authors postulated that this could be due to a combination of decreased postural control and altered upper extremity movements. To the best of our knowledge no previous study has investigated the risk of upper extremity injury following first-diagnosed concussion in college football players.

The purpose of this study is to investigate whether National Collegiate Athletic Association (NCAA) football players are at greater risk of upper extremity injury after their first diagnosed concussion than they were beforehand. It was hypothesized that football players would exhibit an increased risk of upper extremity injury in the year following first-diagnosed concussion when compared to the year prior.

Materials and Methods

Data Acquisition

The Sports Injury Research Archive (SIRA) of the Pacific-12 (Pac-12) athletic conference within the NCAA was queried for de-identified injury data on concussions and upper extremity injuries. The following data are collected into the SIRA for each health event (injury or

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illness) sustained by an athlete: deidentified athlete ID, sex, event year, body part affected, injury/illness type, mechanism of injury, season timing (in season vs off season), event setting, requirement for physician encounter, requirement for surgery, and return-to-play data. Age, position, playing time and whether the player was a starter were not included in the database. Relevant data was extracted and transferred to Microsoft Excel (Version 16.66.1) for analysis. This study received Institutional Review Board (IRB) approval as it utilized previously collected, de-identified data.

Data Collection

The study population involved male collegiate football players who sustained their first diagnosed concussion between January of 2017 and December of 2021. Exclusion criteria were as follows: athletes with non-sport-related concussions, athletes with multiple concussions, and athletes that did not return to previous activity level in their respective sport. Each concussed athlete was then retrospectively followed for the year prior to their concussion as well as one year after their concussion, beginning at 90 days after the date of their concussion. The 90-day grace period was applied to minimize any impact of acute concussion symptoms on the outcomes of the study and focus on the long-term injury risk posed by an athlete sustaining their first diagnosed concussion. In order to maintain a uniform data collection period across all subjects, athletes who did not return within 90 days after their concussion were excluded from the study. Upper extremity injuries were defined as acute injuries of the shoulder, arm, elbow, forearm, wrist, hand, and fingers. Injuries to the shoulder included both those to the gleno-humeral joint and the shoulder girdle and thus encompassed clavicular and scapular fractures. Injuries to the neck, cervical spine, thoracic spine, or chest wall were not included. All upper extremity injuries

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were identified in the year prior to concussion and the year after the concussion, beginning after the 90-day period after the initial event. (**Figure 1**)

Statistical Analysis

An odds ratio was calculated to determine the likelihood of an upper extremity injury occurring within the year after a football player sustained a concussion versus the year before they sustained a concussion. Statistical significance of the results between the concussed groups with and without a subsequent upper extremity injury were defined by a 95% confidence interval (CI). All statistical analysis was performed using Microsoft Excel.

Results

After excluding 36 athletes with repeat concussions and 8 athletes who did not return in the 90 days post-concussion, 160 athletes met inclusion criteria. Of these athletes there were 11 upper extremity injuries sustained in the 365 days prior to the concussion diagnosis along with 26 upper extremity injuries sustained in the 365 days that they were studied after concussion (**Table 1**). The odds of upper extremity injury after concussion were 2.36 (95% CI = 1.13-4.95, p = 0.02) times higher that before concussion. Of the upper extremity locations, shoulder was the most common site of injury with 57.7% of injuries after concussion compared to 19.2% in the hand, 15.4% in the elbow, 7.7% in the forearm, and 0% in the wrist.

On average, upper extremity injuries before concussion occurred 247.3 days (SD 93.5) prior to the concussion. The mean time to injury after concussion was 272.9 days (SD 110.4) (**Table 2**), which correlates to 182.9 days after post-concussion injury data began being collected.

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Discussion

The findings of this study demonstrate that NCAA division I football players from a single conference are at a higher risk of upper extremity injury after their first diagnosed concussion compared to before the concussion. Specifically, the odds of sustaining an upper extremity injury were 2.36 times higher in a one-year period after concussion than in a one-year period prior to concussion. Our results are consistent with those described by Roach et al. in the military population[35] and add substantially to a subject upon which there has been scant research to date.

Our study demonstrated site specific musculoskeletal injury prevalence after concussion of 57.7% in the shoulder, 19.2% in the hand, 15.4% in the elbow, 7.7% in the forearm, and 0% in the wrist. In comparison, Roach et al. found incidence of upper extremity musculoskeletal injury after concussion to be 47.5% in the shoulder, 29.5% in the hand, 16.4% in the elbow, 0% in the forearm and 6.6% in the wrist.[35] Although our findings are not identical, they do suggest a somewhat similar distribution of site specific prevalence for upper extremity musculoskeletal injury after concussion. Specific prevalence differences may be attributable to the relatively small sample size of 26 post-concussion injuries in our study and 61 in Roach et al.[35]

In this study, post-concussion injuries occurred an average of 272.9 days after the inciting concussion, which correlates to almost exactly the midpoint of the collection period (90 – 455 days), accounting for the 90-day buffer period after the concussion. This implies that a prior concussion increases an athlete's risk of upper extremity injury in a long lasting manner and that after 90-days risk of upper extremity injury remains relatively constant. This is supported by research in lower extremity musculoskeletal injury post-concussion which has demonstrated injury risk persists for multiple years after concussion.[15] Furthermore, retrospective studies of

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retired NFL athletes have demonstrated correlations between concussion and future osteoarthritis as well as self-reported head trauma and gait deficiencies.[21,23]

Providing support that future injuries are sequelae of a concussion, and not simply from a playing style more prone to injury overall, is a challenge in studying post-concussion injury. To address this barrier our study examined athletes both before and after their concussion and compared their rate of injury during both time periods. Furthermore, to limit the possibility of acute concussion symptomatology being the etiology of injuries, data was not collected until 90 days after the date of the concussion.

Our findings demonstrate a similar increase in incidence of upper extremity musculoskeletal injury after concussion when compared to previous research showing an increase in sprains, strains, contusions and fractures to the hip, knee, foot and ankle following concussion.[1,2,8-11,13-16,19,20,26,28,37] The first paper to investigate injury rates after concussion was by Nordstrom et al., who demonstrated an approximately 50% increased risk of injury in the first year after concussion.[28] Since then the relationship between injury and concussion has been studied with increasing frequency. Many of the investigations have used NCAA athletes as a study population.[8-11,16,19] Harada et al. demonstrated a dependent relationship between the number of concussions sustained and the risk of future lower extremity injury.[9] In high school athletes, Lynall et al. showed a 34% increase in lower extremity musculoskeletal injury risk after concussion.[20] A recent prospective study by Biese et al. provided further support of lower extremity musculoskeletal injury risk after concussion in the high school population.[2] Furthermore, NFL athletes were shown to have from 2.28 to 2.92 times higher odds of lower extremity musculoskeletal injury after concussion, depending on the number of concussions that

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they had suffered.[1] Notably, we found no studies that investigated the risk of injury for beyond 3 years after concussion, which is an area worthy of future investigation.

The convincing data to support the relationship between concussion and extremity injury has led to a range of research investigating possible explanations.[4,5,17,18,22-24,30] Early work by Martini et al. examining gait mechanics found that those with a history of concussion had slower gait and spent more time in the double leg stance phase of gait than their non-concussed counterparts.[24] Subsequently, the addition of a cognitive task in tandem with gait was shown to widen the gap between those with a history of concussion and those without.[17] Landing biomechanics have exhibited alterations in previously concussed recreational athletes as well as military personnel.[5,18] Another proposed theory for why concussed athletes have increased injury risk is a change in their perception-action coupling behavior; this can alter the way they perceive and respond to their environment.[6,7] It can be supposed that the biomechanical deficits and neurocognitive changes caused by concussion similarly affect the upper and lower extremity musculoskeletal injury risk, but this area deserves further investigation.

Future research should build upon recent work by Mang et al. who investigated newly concussed, symptomatic patients and found them to have impairments in upper extremity rapid bimanual motor tasks.[22] While the work of these authors focuses on acute concussion symptoms, the possibility of a chronic upper extremity coordination deficit after concussion is not unreasonable given the established literature on lower extremity biomechanical deficits, and is worthy of future investigation.

Limitations of this study stem primarily from those inherent to the database it utilized. The database did not include concussion data from prior to the beginning of an athlete's NCAA career. Furthermore, the de-identified data did not include information about playing time or

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position of the athletes and thus insight into an athlete's degree of exposure to injury was limited. The dataset also did not include enough other pertinent, demographic information such as age and starter/reserve status to allow matching with non-concussed controls in any reliable fashion. We were also unable to comment on specific pathology and mechanism of injury; however, all included injuries were musculoskeletal in nature. Finally, it has been well described that NCAA football players experience higher injury rates as their career progresses and as a result our methodology may slightly overestimate the increased risk of upper extremity injury attributable to a prior concussion.[25]

Conclusion

In summary, our data has demonstrated a link between first-time concussions in NCAA football players and an increased risk of subsequent upper extremity injury. Thus, it is important for athletes and athletic trainers to be wary of the long-term risk for upper extremity musculoskeletal injury posed by concussion and take steps to address its impact in rehabilitation and injury recovery protocols. Further characterization of this risk as well as delineating its neurocognitive and biomechanical basis represent important areas of further research.

Disclosure of interest:

The authors report no conflicts of interest

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Figure 1: Timeline of injury collection data

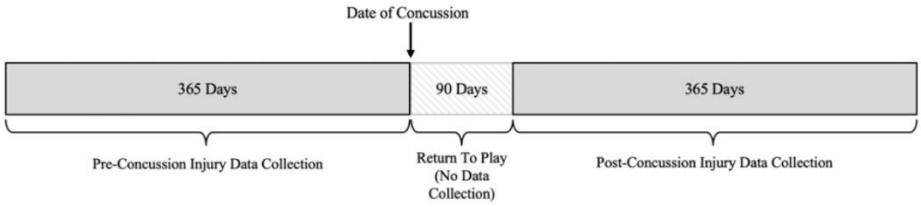

Table 1: Stratification of upper extremity injuries before and after concussion by body part.

Table 1. Upper extremity injuries before and after concussion.

	Before Concussion	After Concussion	OR (95% CI)	P-value
Total # UE Injuries	11	26	2.36 (1.13-4.95)	**0.02**
Shoulder (%)	7 (63.6)	15 (57.7)	2.14 (0.85-5.40)	0.11
Elbow (%)	1 (9.1)	4 (15.4)	4.00 (0.44-36.2)	0.22
Forearm (%)	1 (9.1)	2 (7.7)	2.00 (0.18-22.3)	0.58
Wrist (%)	1 (9.1)	0 (0)	N/A	N/A
Hand/Fingers/Thumb (%)	1 (9.1)	5 (19.2)	N/A	N/A

Table 2: Timing of upper extremity injuries relative to player being concussed

Table 2. Timing of Upper Extremity Injuries Before and After Concussion

	Before Concussion	After Concussion
Mean Time (days) Between Concussion and UE Injury (SD)	-247.3 (93.5)	272.9 (110.4)
Season in which UE Injuries Occurred (%)		
In-Season (Includes pre/post-season)	6 (54.5)	18 (69.2)
Off-Season	5 (45.5)	8 (30.8)

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